Experiment - 14

Student Name: Aditya Pratap Singh
Branch: CSE - AIML
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Subject Name: ADBMS

UID: 24BAI70438
Section/Group: 24AIT_KRG-1_B
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Aim - To study Amazon RDS, create a database instance on AWS RDS, understand security groups, and connect a cloud RDS database to a local database client such as pgAdmin.

Software used –

- AWS Management Console / AWS account
- Amazon RDS
- Amazon EC2 (for the MySQL-client approach shown in the guidelines)
- pgAdmin (for the PostgreSQL local-connection approach)
- Internet connection

Theory - Amazon RDS is a managed relational database service provided by AWS. The guidelines focus on scalability, performance, advanced security, costeffectiveness, manageability, availability and durability.

The experiment contrasts databases running on EC2 with RDS. Running a database on EC2 can provide OS access, advanced DB option tuning, vendor-specific requirements and versions not provided by AWS, but it also introduces administration, backup/DR, monitoring and setup overhead.

Features –

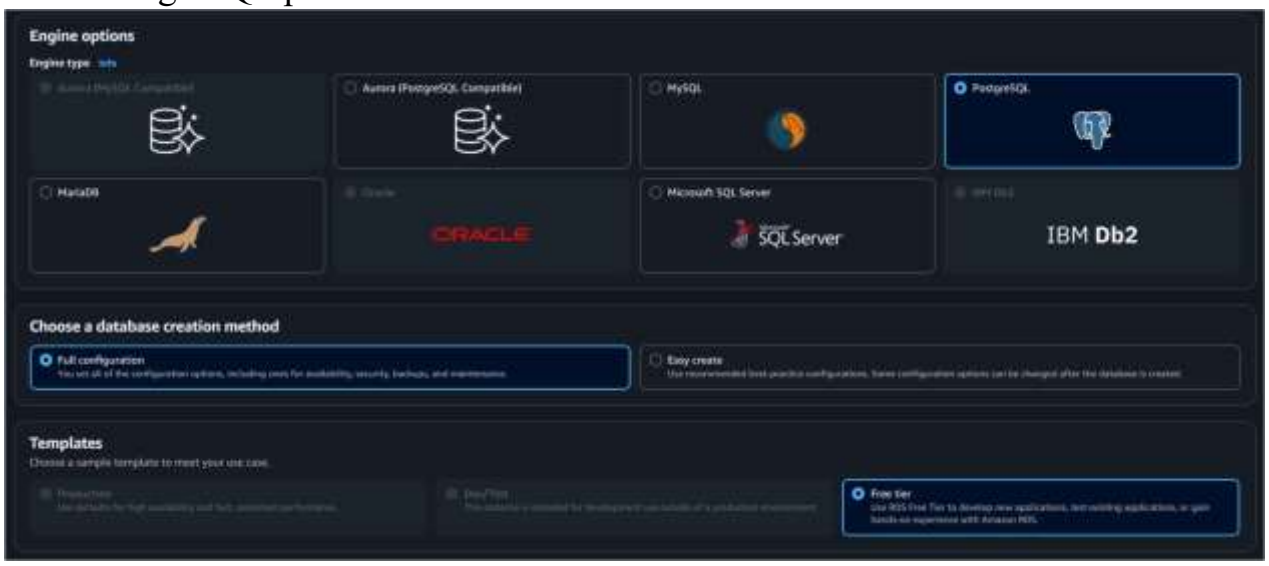
Feature	Purpose
CloudWatch	Performance and health monitoring
Automated Backups	Disaster recovery
Manual Snapshots	User-created backups / long-term retention
Multi-AZ Deployment	High availability using a standby replica in another AZ
Read Replicas	Read-only copies for load balancing
Security (IAM, KMS, SSL)	Data protection, access control, compliance and safety
Performance Insights	SQL/load analysis and performance tuning
Storage Auto Scaling	Automatically grows storage as needed
Enhanced Monitoring	OS-level real-time metrics
Cross-Region Replicas	Replication for global availability
Parameter / Option Groups	Database configuration controls

Steps to create and configure the DB –

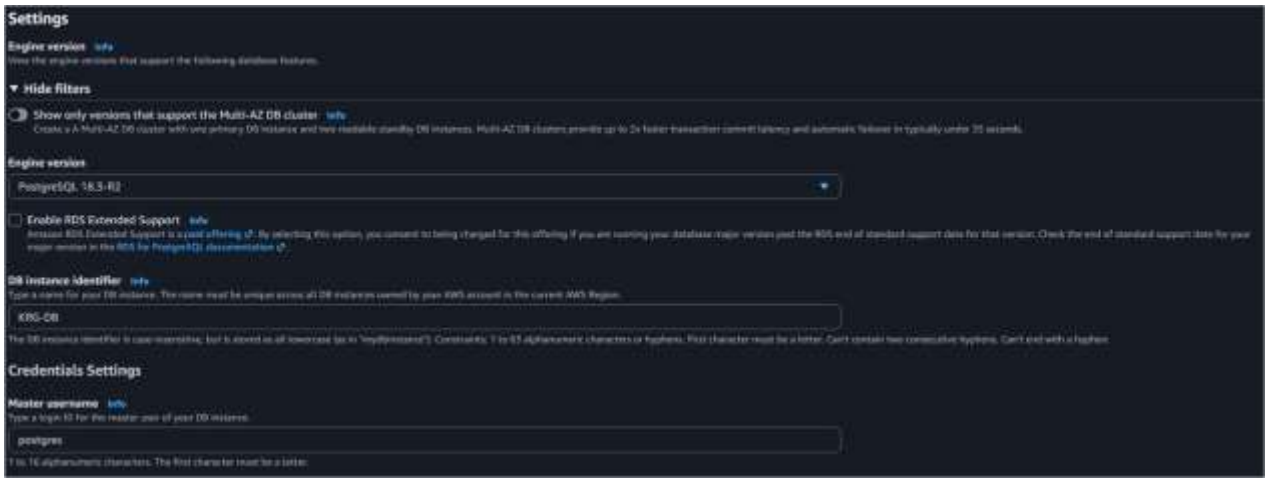
Go to AWS RDS Homepage:



Select PostgreSQL provider:



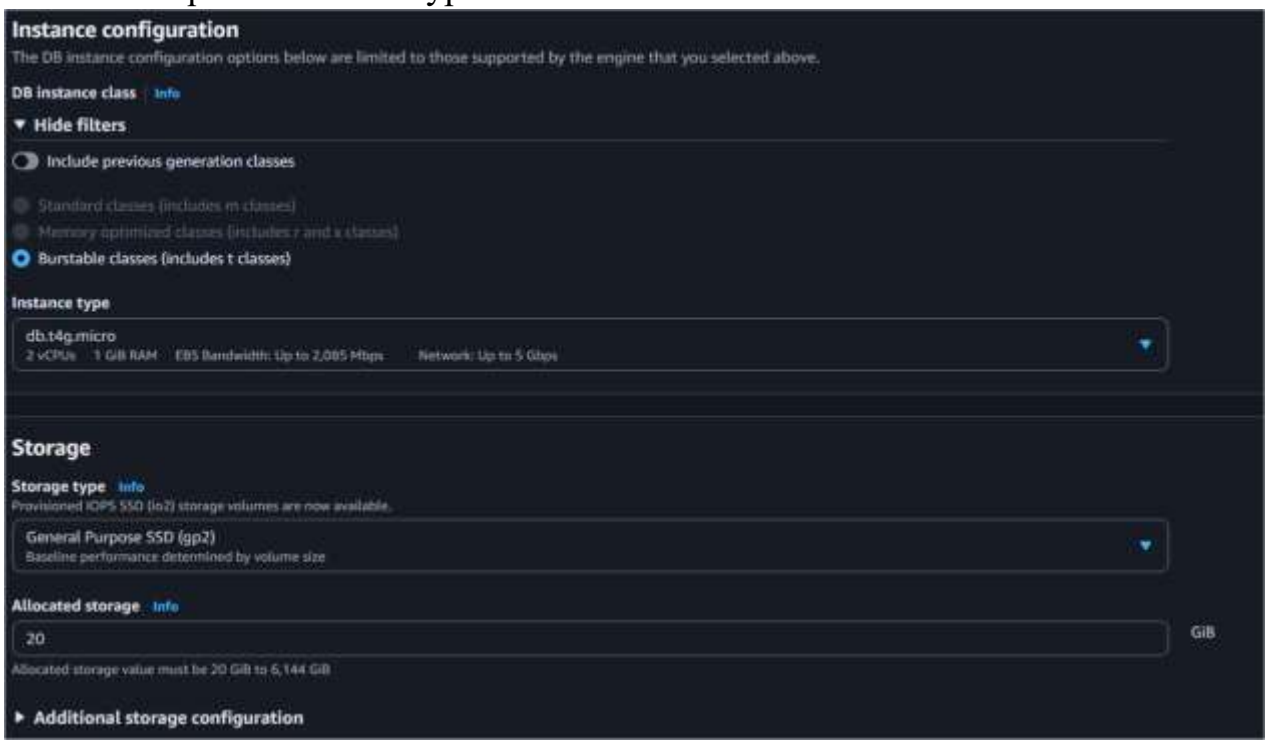
Set DB Instance Identifier:



Select Slef-Managed and set Master Password



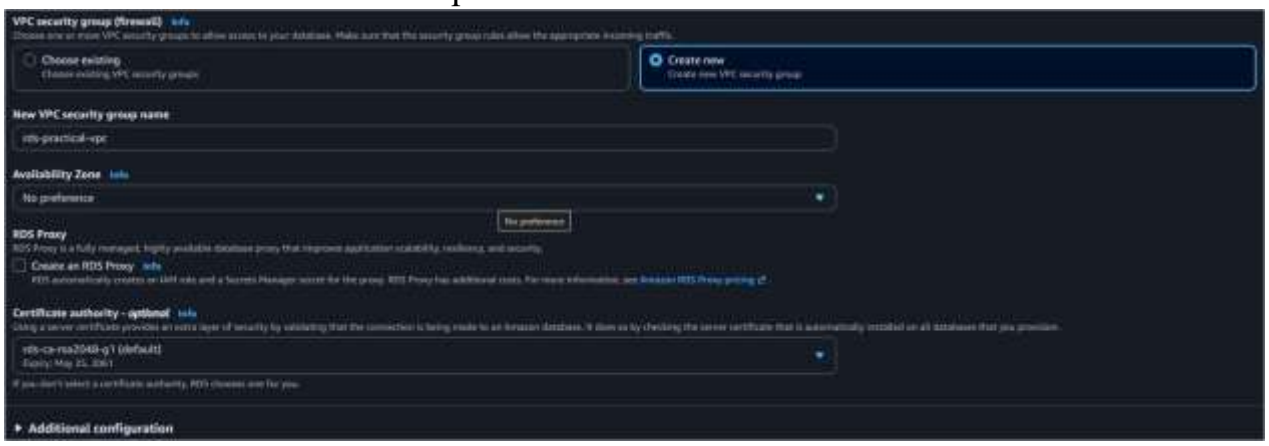
Select the required Instance type:



Allow Public access:



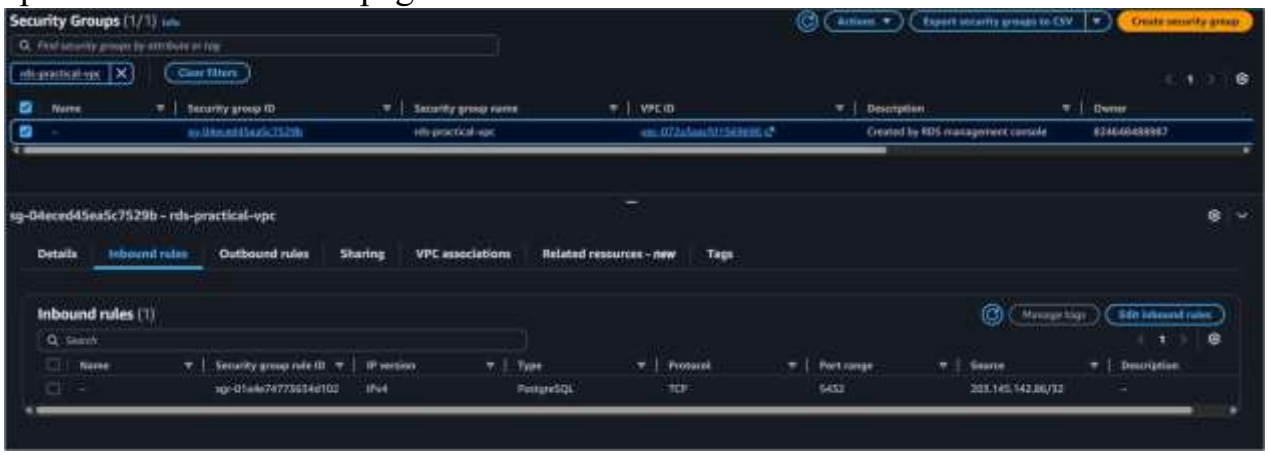
Create custom VPC rules if required:



Wait for the DB to start



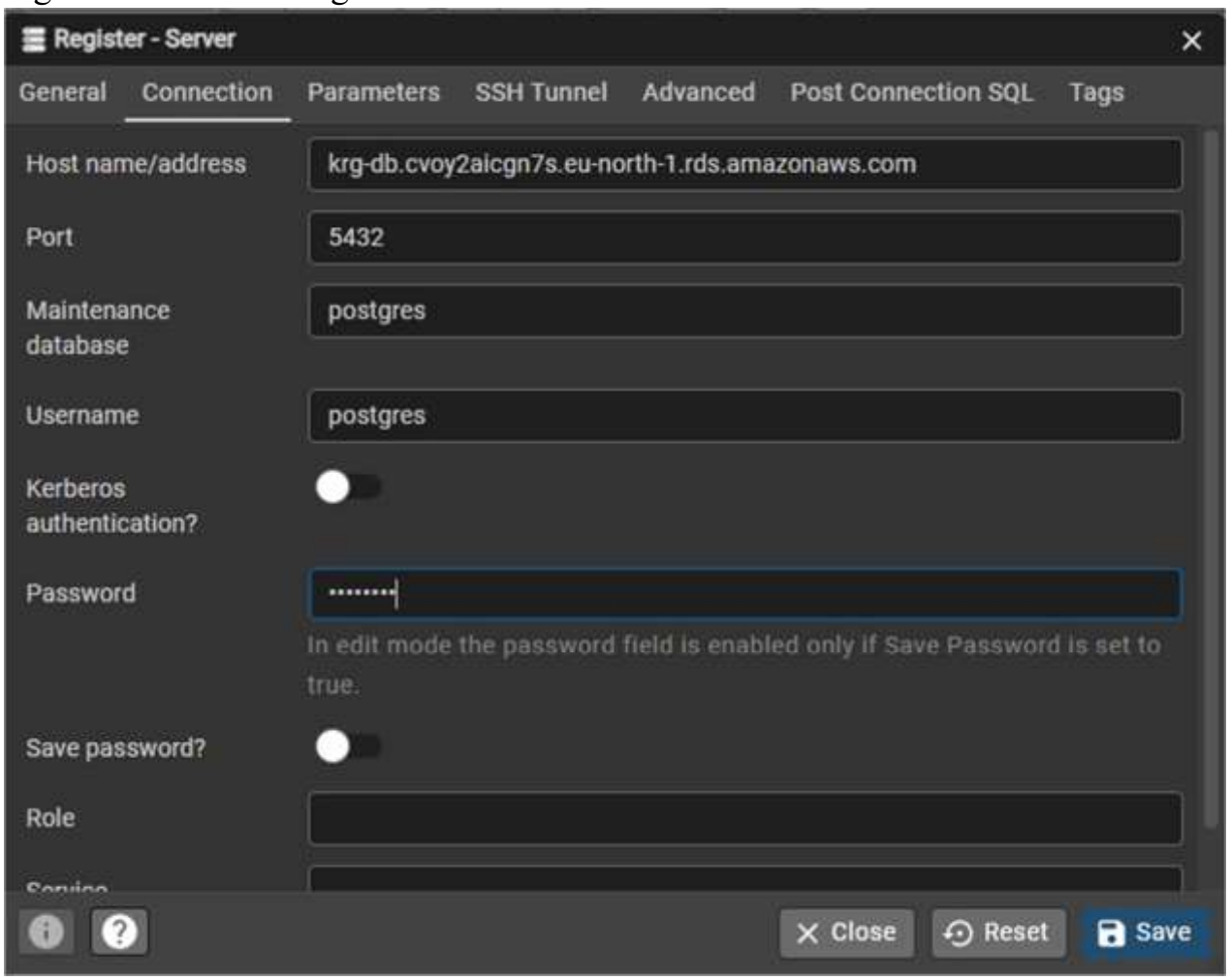
Open the InBound rules page:



Add the host IP address



Register the server in PgAdmin:



The DB is added successfully:



Verify the connection:

